STAT 231: Blog Project

Timeline

All deliverables must be submitted by 9:00 PM (unless otherwise indicated) on the dates provided at the appropriate submission location. Adherence to these deadlines is required to help us keep on pace for the semester.

Activity	Date	Submission
Initial Proposal	Thursday, Oct. 15	New GitHub Issue (Blog plan)
Revised Proposal	Tuesday, Oct. 20	New comment on GH Issue (Blog plan)
Wrangled dataset(s) and code	Monday, Nov 9	your GitHub project repo
Blog Post	Tuesday, Dec. 1 by 5pm	Website (using GitHub Pages)
1-on-1 presentation + Q&A	Week of Monday, Nov. 30	My office
Peer share	Tuesday, Dec. 8	In class
Reflection	Wednesday, Dec. 9 by 5pm	Gradescope

Assessment

The project components will carry the following weight:

Component	Points
Proposal(s)	5
Wrangled dataset(s) and code	5
Blog post	10
1-on-1 presentation + Q&A	75
Peer share	5

Proposals (5 pts per proposal, with score averaged across the 2 proposals)

Initial proposal

- ☐ To facilitate timely feedback, the initial proposal is submitted by its due date.
- ☐ To facilitate assessment, the initial proposal is submitted as a new issue to your group blog project repo within our GitHub classroom organization.
- ☐ The issue title includes "Proposal" in the name.
- ☐ The issue body includes all four prompts and corresponding responses are clear and complete.

Revised proposal

- ☐ To facilitate timely feedback, the revised proposal is submitted by its due date.
- ☐ To facilitate assessment, revisions to the proposal are provided as a response to my response to the original issue (the original issue is *not* edited!).
- ☐ If requests for revisions are received, feedback given is addressed in full.

Wrangled dataset(s) and code (5 points)

Organization

- ☐ Files are logically organized within folders.
- ☐ All files necessary for reproducing your project thus far are on GitHub (qmd, datasets, other imported files, …).

Functionality and code choice

- ☐ Code outputs a permanent, clean dataset that will be imported into your index.qmd file
- ☐ Code is fully reproducible without modification. Filepaths are relative to the location of the R or qmd file.
- ☐ Code runs without error and correctly produces desired output.
- ☐ Code is efficient with little or no redundancies in processes or stored objects.
- ☐ Code demonstrates proficiency with packages learned in class (primarily **tidyverse** and **tidyverse**-friendly functions). Code does not resort to base R.

Readability and style

- ☐ Use of indentation, white space between sequences of code, and separate code chunks promotes readability.
- ☐ Code follows recommendations for placement of line breaks (after pipe operators, between plot layers, logical breaks within functions if line is 80+ characters).
- ☐ Object and function names make the purpose or content clear.
- ☐ Code chunks are split logically and located in relevant places in the document.
- ☐ Code is well-documented with the use of code comments. Documentation explains the *why* of the code.
- ☐ Code is well-organized and easy to follow.
- ☐ Only code that is evaluated is included in the final submission.

Blog Post

File management

- ☐ Files are logically organized within folders in your blog repo.
- ☐ Folder and file names follow file-naming expectations (no spaces or special characters except "-" or "_") and make the content or purpose clear. It would be easy for anyone to navigate the contents of the repo without additional context.
- ☐ All files necessary for reproducing your project are on GitHub (qmd, PDF, Rproj, datasets, other imported files, …).
- ☐ There are no extra copies of files, "messy"/scratchwork files, or other files not used for your final project.
- ☐ .gitignore is used effectively to ignore unnecessary files (e.g., .DS_Store)

Code

Readability: Code is formatted in a consistent style (consistent use of spacing, indentation, and naming conventions).

- ☐ Code follows recommendations for spacing (around operators, between arguments, etc.)
- ☐ Code follows recommendations for placement of line breaks (after pipe operators, between plot layers, logical breaks within functions if line is 80+ characters otherwise).
- ☐ Use of indentation, white space between sequences of code, and separate code chunks promotes readability.
- ☐ Object and function names make the purpose or content clear.

Documentation: Code is well-organized. Style, organization, and documentation make the code easy to follow. Location of code chunks is relevant to document section.

- ☐ Code chunks are split logically and located in relevant places in the document.
- ☐ Code is well-documented with the use of code comments. Documentation explains the *why* of the code.
- ☐ Code is well-organized and easy to follow.
- ☐ Only code that is evaluated is included in the final submission.

Functionality and code choice

- ☐ Code is fully reproducible without modification. Filepaths are relative to the location of the R or qmd file.
- ☐ Code chunk options are used correctly and customized as appropriate.
- ☐ Code runs without error and correctly produces desired output.
- ☐ Code is efficient with little or no redundancies in processes or stored objects.
- ☐ Code demonstrates proficiency with packages learned in class (primarily **tidyverse** and **tidyverse**-friendly functions)
- ☐ No (or only minimally necessary) data wrangling is present within any webpage's *.qmd* file.

Content

For any statistical methods implemented:

- ☐ Choice of statistical method is appropriate and carried out correctly.
- ☐ Communication of statistical methods and results is correct.
- ☐ Communication of statistical methods and results is clear (easy for a reader to follow and understand).
- ☐ Communication of statistical methods and results is appropriate to audience level.

For any data visualizations:

- ☐ Visualizations are relevant and use effective visual cues.
- ☐ Graphs are well-suited to the type and scale of data they represent.
- ☐ Visualizations are easy to read, with clear labels and legends.
- ☐ Width and height of figures are aesthetically pleasing.
- ☐ Interactivity, if used, enhances the chosen representation of the data and is user-friendly.

For any tables:

- ☐ Tables are relevant and appropriate.
- ☐ Tables complement visualizations and are not redundant.
- ☐ Tables are publication quality, easy to read, with clear labels and nice formatting
- ☐ Number of digits displayed for numeric columns are consistent and appropriate.
- ☐ Interactivity, if used, enhances the chosen representation of the data and is user-friendly.

References:

- ☐ Data sources and any other relevant sources are appropriately cited.
- ☐ All necessary references appear to be included, and each reference list item is as complete as possible (e.g., title, author, date, link).

Other content:

- ☐ Blog incorporates at least one of the methods introduced in the second half of the semester: text analysis, network science, unsupervised learning, and/or spatial data.
- ☐ Purpose of blog is clear and elements are cohesive.
- ☐ Blog is compelling (interesting, creative, thought-provoking, or meaningful) with a solid introduction.
- ☐ Data sources and context are clearly stated, including, but not limited to: size of the data, how/when it was acquired by the blog team, how/when the data was originally acquired if from a survey, experiment, or other study. Relevant variables are explained.
- ☐ Claims or conclusions are adequately supported.
- ☐ Language of the blog is appropriate to the audience and clear. Context and description of the topic is sufficient for a reader without subject-matter expertise to follow along.
- ☐ Little or no grammatical errors or spelling mistakes.

Formatting:

- ☐ Site is well-organized and uses headings and subheadings as appropriate to make the document easy to navigate.
- ☐ Text and analysis is effectively interwoven.
- ☐ Good choice of when to display code. Any code displayed is relevant to the audience.

1-on-1 Presentation + Q&A

Each 5 minute section is worth 20 points.

- ☐ Shows up on time for meeting.
- ☐ Presentation is between 4-5 minutes
- ☐ In presentation, introduces topic/research question(s) clearly.
- ☐ In presentation, explains data acquisition process clearly.
- ☐ In presentation, interprets relevant visualization and table components accurately.

- ☐ In presentation, discusses appropriate conclusions.
- ☐ In Q&A, responds fully, accurately, and in detail to the coding questions.
- ☐ In Q&A, responds fully, accurately, and in detail to the conceptual questions.

Peer Share

- ☐ Comes to class prepared to participate in the peer share.
- ☐ Thoughtfully reads and explores the two assigned blog projects.
- ☐ Engages actively in discussion with peers, including asking questions and responding to others' ideas.
- ☐ Identifies something interesting, effective, or memorable about others' work.
- ☐ Identifies a potential extension, next step, or new question inspired by the projects.
- ☐ Reflects on something learned from others' projects that could inform their own future data science work.

Reflection

- ☐ Reflection is complete, thoughtful, and submitted on time.